

NATIONAL INSTITUTE OF TECHNOLOGY ANDHRA PRADESH

Department of Electronics and Communication Engineering



Design and Implementation of a Multi-Band Analog Audio Equalizer

PROJECT REPORT

Minor Project II (EC399)

Presented by:

Mohit Kumar Gupta [623148]

Satyam Kumar [623172]

Vishal Ray [622271]

Guide:

Dr. M. C. Raju

Assistant Professor

NIT Andhra Pradesh



CERTIFICATE

This is to certify that the project report entitled "Design and Implementation of a Multi-Band Analog Audio Equalizer" is a Bonafide record of the work carried out by Mohit Kumar Gupta (623148), Satyam Kumar (623172), and Vishal Ray (622271) in fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering at the National Institute of Technology Andhra Pradesh.

This work has been carried out under my supervision during the academic year 2025–2026.

Guide:

Dr. M. C. Raju
Assistant Professor
DECE
NIT Andhra Pradesh

Dr. B. Narasimha Rao
Head of Department
DECE
NIT Andhra Pradesh



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ABSTRACT

This project presents the design and implementation of a 5-band analog parametric audio equalizer covering the full audio frequency range from 20 Hz to 20 kHz. The system is based on the State Variable Filter (SVF) topology, enabling independent control of center frequency, gain, and bandwidth (Q-factor) for each band.

The equalizer consists of three tunable peaking filters centered at 250 Hz, 800 Hz, and 2.5 kHz, along with low and high shelving filters for bass and treble control. The design uses TL074 JFET-input operational amplifiers to achieve low noise, high input impedance, and adequate bandwidth for audio applications.

A complete signal chain is developed, including input/output interfacing through 3.5 mm TRRS audio jacks, ESD protection using TVS diodes, and proper impedance matching. The circuit is designed and verified using LTspice and Proteus simulations, followed by schematic capture and PCB layout using KiCad.

The results demonstrate accurate frequency response, ± 9 dB gain control, and stable performance across all bands. The project highlights the practical implementation of analog filter design concepts and provides a foundation for further development into a fully functional audio processing device.



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1. Introduction

1.1 Background and Motivation

Audio equalization is the process of selectively adjusting the amplitude of specific frequency bands within an audio signal. It is fundamental to professional audio production, live sound reinforcement, acoustic correction, and consumer hi-fi systems. Without equalization, audio systems suffer from uneven frequency response introduced by room acoustics, speaker colorations, and source limitations — all of which degrade the perceived quality of sound.

Digital equalizers have become dominant in modern audio processing owing to their flexibility and precision. However, analog equalizers remain indispensable in applications demanding zero latency, simplicity of hardware, and the characteristic warmth that continuous-time processing preserves. For educational purposes, an analog parametric equalizer also serves as a comprehensive vehicle for applying filter theory, op-amp circuit design, and frequency-domain analysis to a real-world problem.

1.2 Project Objectives

- Design a 5-band analog parametric equalizer covering the full audio spectrum (20 Hz – 20 kHz).
- Implement three fully tunable SVF-based peaking bands at 250 Hz, 800 Hz, and 2.5 kHz with ± 9 dB gain and adjustable Q.
- Implement low shelving and high shelving filters with tunable corner frequencies and ± 9 dB gain.
- Design a complete signal chain including 3.5 mm TRRS stereo input/output jacks with impedance matching and ESD protection.
- Capture and verify the full schematic in KiCad EESchema and complete a 4-layer PCB layout in KiCad PCBnew.
- Verify all filter blocks through LTspice AC sweep simulation (20 Hz – 20 kHz) and Proteus simulation using exact TL074C SPICE models.

1.3 Scope of This Report

This report presents the complete design and implementation work carried out over the full duration of the Minor Project II. It covers the theoretical foundation, all circuit design decisions, simulation results, schematic and PCB design, and verification outcomes. It is structured as a single comprehensive document encompassing all work from initial architecture selection through final PCB layout.

2. Background and Theory

2.1 Types of Equalizers

Three principal equalizer architectures exist in analog audio design:

- Graphic Equalizer: Fixed center frequencies per band, typically 10 or 31 bands. Simple to operate visually using sliders but offers limited precision and fixed bandwidth.
- Parametric Equalizer: Adjustable center frequency (f_0), adjustable Q (bandwidth), and adjustable gain per band. Provides the highest precision and is standard in professional studio applications. This project implements this type.



- Shelving Equalizer: Boosts or cuts all frequencies above (high shelf) or below (low shelf) a corner frequency f_c . Used for bass and treble tone control. Implemented in this project as the two end bands.

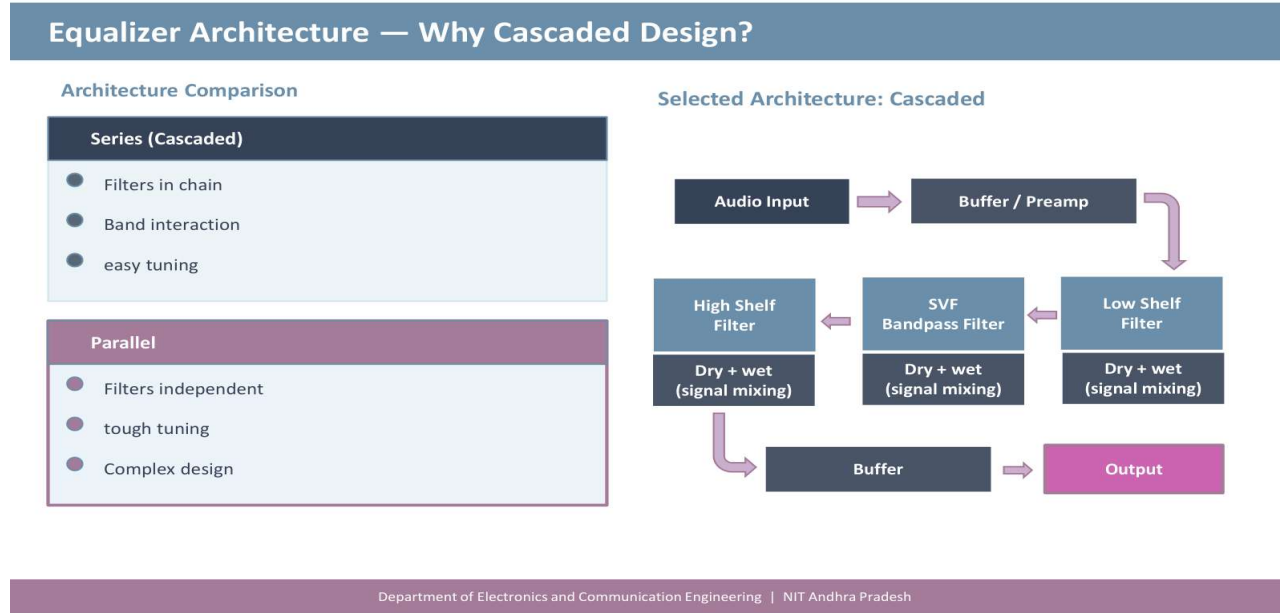


Figure 2.1 — Cascaded equalizer architecture adopted in this project

2.2 Analog vs Digital Equalizers

Feature	Analog Equalizer	Digital Equalizer
Signal Domain	Continuous-time	Discrete-time (sampled)
Latency	Zero latency	Processing delay present
Noise	Thermal & op-amp noise	Quantization noise
Flexibility	Fixed hardware design	Software reprogrammable
Power	Low	Higher (DSP core required)
Audio Quality	Warm, continuous-time character	Precise, bit-perfect control

2.3 Human Hearing and Band Selection

The human auditory system perceives pitch logarithmically and exhibits frequency-dependent loudness sensitivity (Fletcher-Munson curves). The three peaking bands in this equalizer were positioned at 250 Hz, 800 Hz, and 2.5 kHz — separated by approximately two octaves each — based on their psychoacoustic significance:

- 250 Hz (Bass band): Warmth and body of bass guitar, kick drum fundamental, male vocal chest resonance.



- 800 Hz (Midrange band): Male and female vocal presence, snare body, guitar midrange character.
- 2.5 kHz (Upper midrange band): Vocal intelligibility, guitar pick attack, piano clarity and definition.

Two-octave spacing between bands minimises inter-band interaction, ensuring that adjusting one band does not significantly affect the response of adjacent bands.

3. System Architecture

3.1 Overall Signal Chain

A cascaded (series) architecture was selected over a parallel arrangement. In a cascaded design, each filter block processes the output of the previous one, which simplifies gain staging and makes individual band tuning straightforward. The complete signal path is:

```
3.5 mm TRRS Input Jack (J1) → DC Blocking + Impedance Matching → ESD Protection (U1)
→ Low Shelf Filter (fc: 120-500 Hz, Gain: ±9 dB)
→ SVF Peaking Band 1 (f0: 120-500 Hz, Q: adjustable, Gain: ±9 dB)
→ SVF Peaking Band 2 (f0: 400-1600 Hz, Q: adjustable, Gain: ±9 dB)
→ SVF Peaking Band 3 (f0: 1.2-5 kHz, Q: adjustable, Gain: ±9 dB)
→ High Shelf Filter (fc: 1.2-5 kHz, Gain: ±9 dB)
→ Output Buffer (R4=100Ω) → ESD Protection (U2) → 3.5 mm TRRS Output Jack (J2)
```

3.2 Op-Amp Selection — TL074CDR

The TL074CDR (Texas Instruments) quad JFET-input op-amp was selected as the core active element for all filter stages. Five ICs provide 20 op-amp units for the entire signal chain.

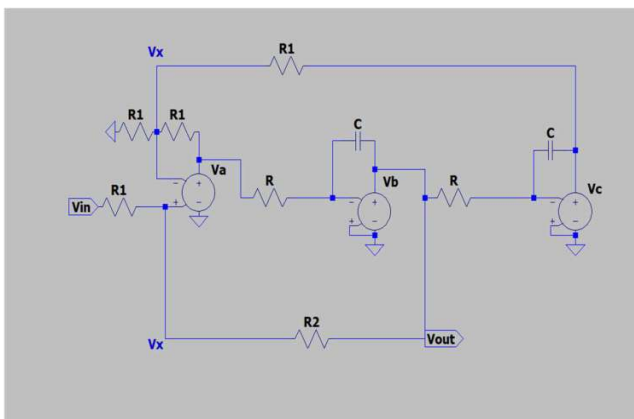
Parameter	TL074CDR Value	Significance
Supply voltage	±15 V	Full ±13 V output swing for audio levels
Slew rate	13 V/μs	Adequate for 20 kHz full-scale signal
GBW product	3 MHz	Sufficient gain margin at audio frequencies
Input bias current	65 pA	Negligible DC error across RC filter networks
Noise voltage	18 nV/√Hz	Low noise floor suitable for audio
Input offset voltage	3 mV	Minimal DC offset at filter outputs
Package	SOIC-14 (CDR)	Used in PCB layout; compact SMD footprint
Channels per IC	4	5 ICs = 20 op-amp stages total

4. Filter Design and Analysis

4.1 State Variable Filter (SVF) — Topology

All five filter blocks are built upon the KHN (Kerwin-Huelsman-Newcomb) State Variable Filter topology. This structure uses a summing amplifier followed by two integrators connected in a feedback loop, yielding simultaneous low-pass (LP), band-pass (BP), and high-pass (HP) outputs. The critical advantage for parametric equalizer design is that the three parameters — center frequency (ω_0), quality factor (Q), and gain — can be adjusted independently without cross-coupling.

SVF Mathematical Model



State Variable Filter

Bandpass Transfer Function

$$H(s) = \frac{(\omega_0/Q) \cdot s}{s^2 + (\omega_0/Q) \cdot s + \omega_0^2}$$

Key Parameters:

ω_0	$= 1 / (R \times C)$	Center angular frequency — set by integrator R and C
Q	$= 1/3(1-K)$	Quality factor — controls bandwidth (independent of gain)
f_0	$= \omega_0 / (2\pi)$	Center frequency in Hz
Dc Gain	$= 3K$	Unity peak when Q resistor is in BP feedback path
K	$= R2/(R1+R2)$	

Figure 4.1 — State Variable Filter topology and mathematical model

Parameter	Formula	Controlled By
Center frequency ω_0	$1 / (R \times C)$	Dual-gang potentiometer on both integrators
Quality factor Q	$1 / [3(1 - K)]$ where $K = R_a / (R_a + R_b)$	Single Q-control potentiometer
Bandpass peak gain	3K	Same potentiometer as Q control (decoupled)
Shelf transition freq.	$1 / (R_{\text{pot}} \times C)$	Single potentiometer per shelf block

4.1.1 Q–Gain Decoupling — Key Design Achievement

In a standard SVF, the bandpass output peak amplitude equals $Q \times V_{in}$. This means that increasing Q to narrow the bandwidth simultaneously increases the boost magnitude — an unacceptable coupling in a parametric equalizer. This problem was solved by repositioning the Q -control resistor into the BP output feedback path of the summing amplifier rather than within the SVF loop itself. With this modification, the peak amplitude remains constant at V_{in} regardless of the Q setting, achieving fully independent Q and gain control.

4.1.2 Dual-Gang Frequency Control

The center frequency $\omega_0 = 1/(RC)$ requires that both integrators in the SVF track exactly the same R value. Using two independent potentiometers would make simultaneous matching impossible in practice. Dual-gang potentiometers (one shaft driving two identical wiper tracks) ensure that both integrators always see equal resistance values, maintaining perfect SVF symmetry and a clean, undistorted filter response at any frequency setting.

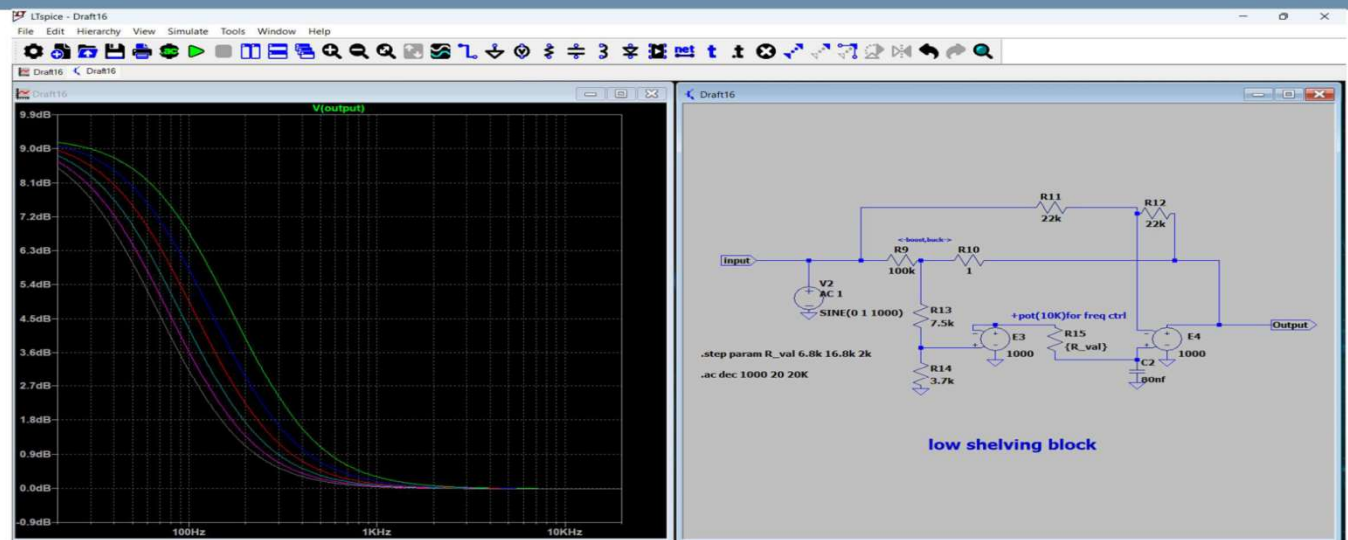
4.2 Low Shelving Filter

The low shelving filter boosts or cuts all frequencies below its adjustable corner frequency f_c . Architecturally, the SVF LP output is scaled by gain factor K and summed with the unfiltered dry signal. The resulting transfer function is:

$$H(s) = \alpha + K \cdot \left[\frac{1}{(SRC)} \right] \quad \omega_0 = \frac{1}{R_{pot} \cdot C11}$$

Behaviour: at $f \ll f_c$ the LP output tracks the input and total gain = $\alpha + K$ (boost/cut region). At $f \gg f_c$ the LP output collapses to zero and gain = α (flat region). Setting $K > 0$ via the gain pot gives boost; $K < 0$ gives cut. The corner frequency is continuously tunable from approximately 120 Hz to 500 Hz.

Block Analysis — Low Shelf Filter (frequency control)

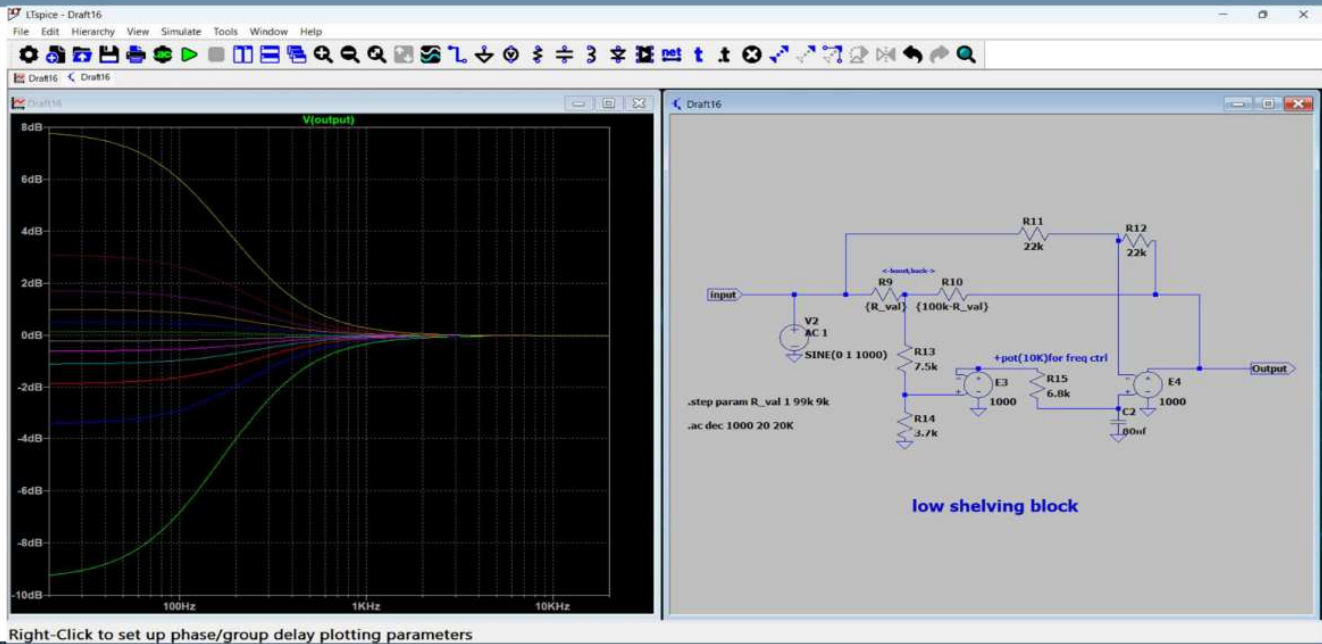


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Figure 4.2 — Low shelf filter: LTspice AC sweep showing corner frequency shift with RV2



Block Analysis — Low Shelf Filter (Gain control)



Right-Click to set up phase/group delay plotting parameters

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Figure 4.3 — Low shelf filter: LTspice AC sweep showing gain sweep from -9 dB to +9 dB.

4.3 SVF Peaking Bandpass Filter — Band 1 (250 Hz)

The 250 Hz peaking band uses the SVF bandpass output mixed back with the dry signal. The standard second-order bandpass transfer function is:

$$H_{BP(s)} = \left(\frac{\omega}{Q}\right) \cdot \frac{s}{\left[s^2 + \left(\frac{\omega}{Q}\right) \cdot s + \omega^2\right]}$$

This band targets the bass-warmth region of the audio spectrum and covers a tuning range of 120 Hz to 500 Hz via dual-gang potentiometers RV3/RV4. Integrator capacitors C3 and C4 (33 nF each) set the base RC time constant.

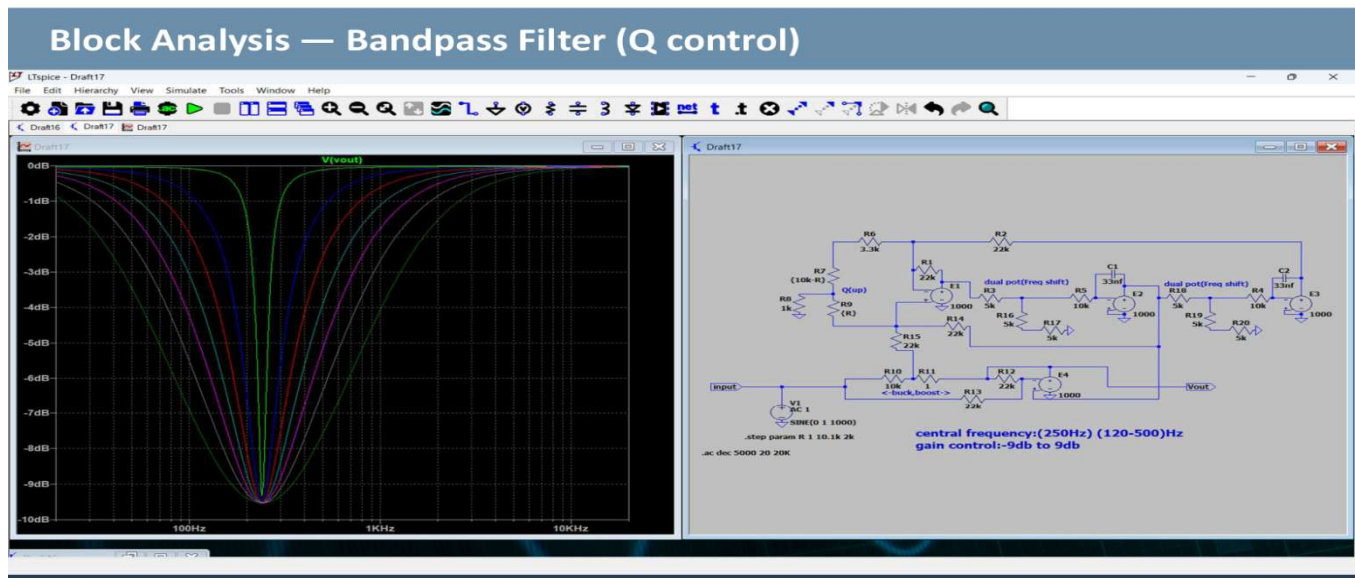


Figure 4.4 — Band 1 (250 Hz): Q control showing bandwidth variation (LTspice)

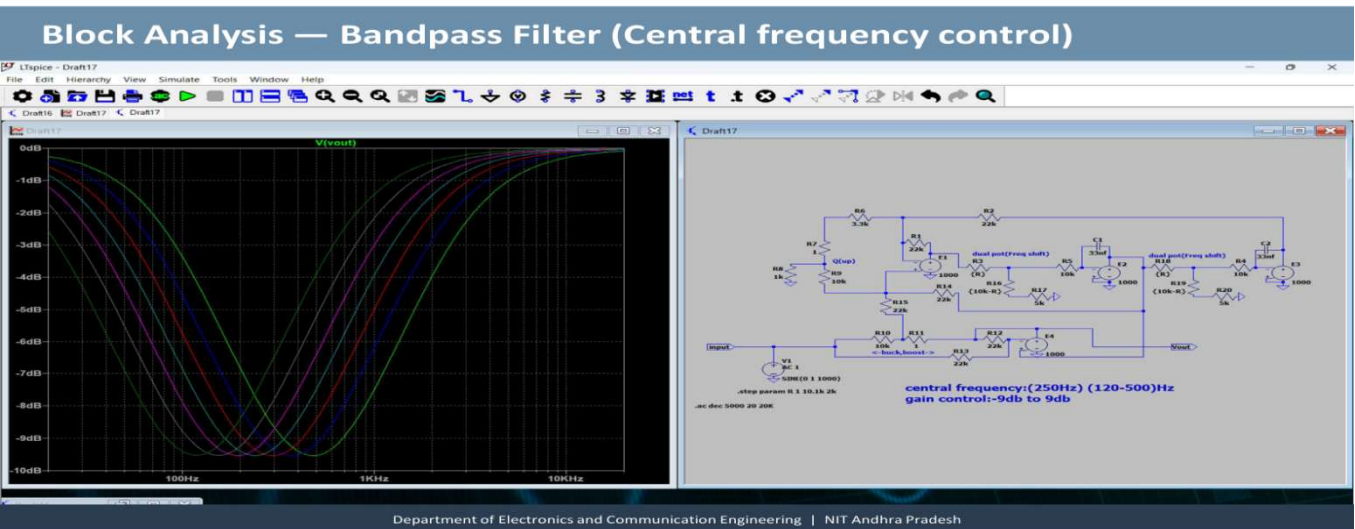


Figure 4.5 — Band 1 (250 Hz): Center frequency shift via dual-gang pot (LTspice)

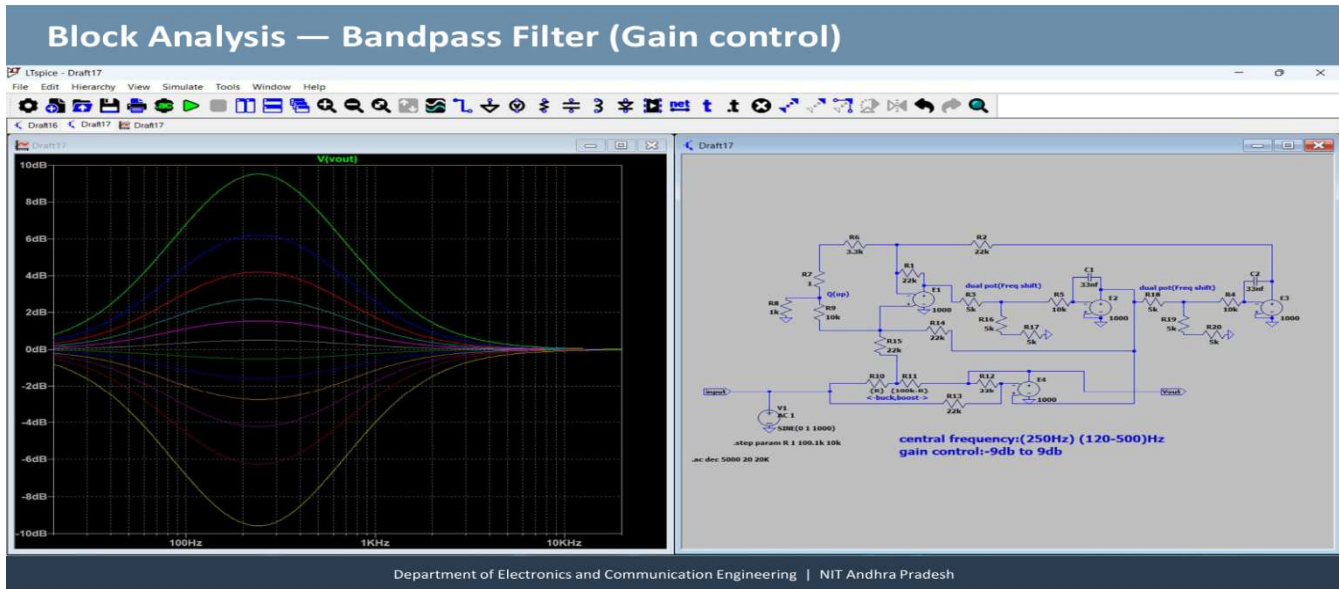


Figure 4.6 — Band 1 (250 Hz): Gain sweep from -9 dB to +9 dB (LTspice)

4.4 SVF Peaking Bandpass Filter — Band 2 (800 Hz) and Band 3 (2.5 kHz)

Bands 2 and 3 follow identical topology to Band 1, with component values scaled to achieve the target center frequencies. The 2-octave spacing between bands (250 Hz → 800 Hz → 2.5 kHz) ensures minimal interaction between adjacent bands.

Band	Center Freq.	Tuning Range	Integrator Caps	Freq. Pot	Gain Pot
Band 1 — Bass	250 Hz	120–500 Hz	C3, C4 = 33 nF	RV3 / RV4 (dual)	RV5
Band 2 — Midrange	800 Hz	400–1600 Hz	C7, C8 = 10 nF	RV7 / RV9 (dual)	RV8
Band 3 — Upper Mid	2.5 kHz	1.2–5 kHz	C9, C10 = 3.3 nF	RV10 / RV12 (dual)	RV11

4.5 High Shelving Filter

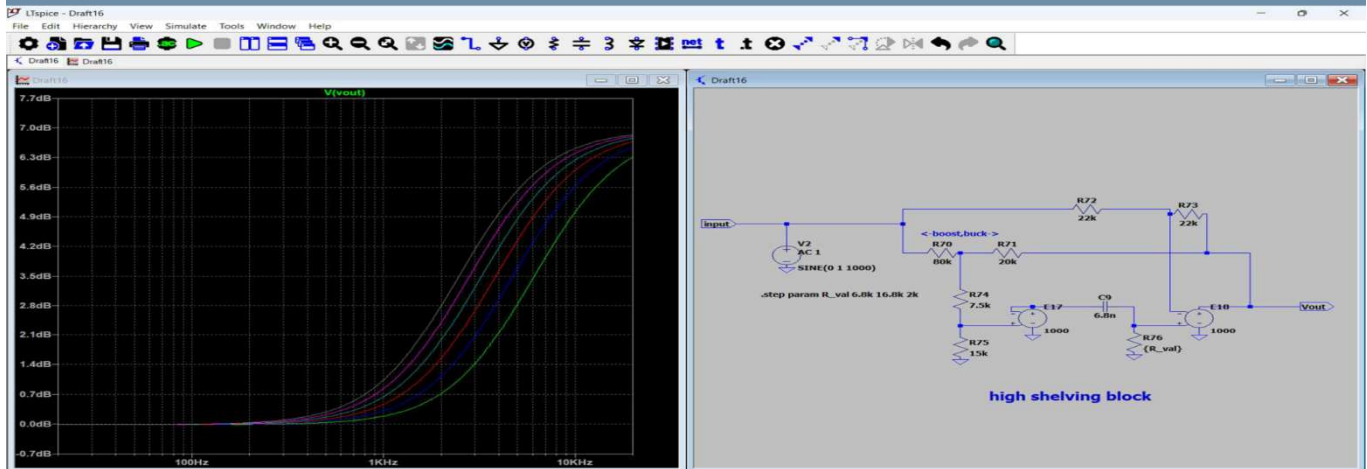
The high shelving filter is the dual of the low shelf. The SVF HP output is scaled and summed with the dry signal. The transfer function is:

$$H(s) = \alpha + K \cdot \frac{sRC}{1+sR} \quad \omega_0 = \frac{1}{R_{pot} \cdot C2}$$

At $f \gg f_c$ the HP output tracks the input (boost/cut region). At $f \ll f_c$ the HP output collapses (flat region). Corner frequency is tuneable from 1.2 kHz to 5 kHz via RV15.



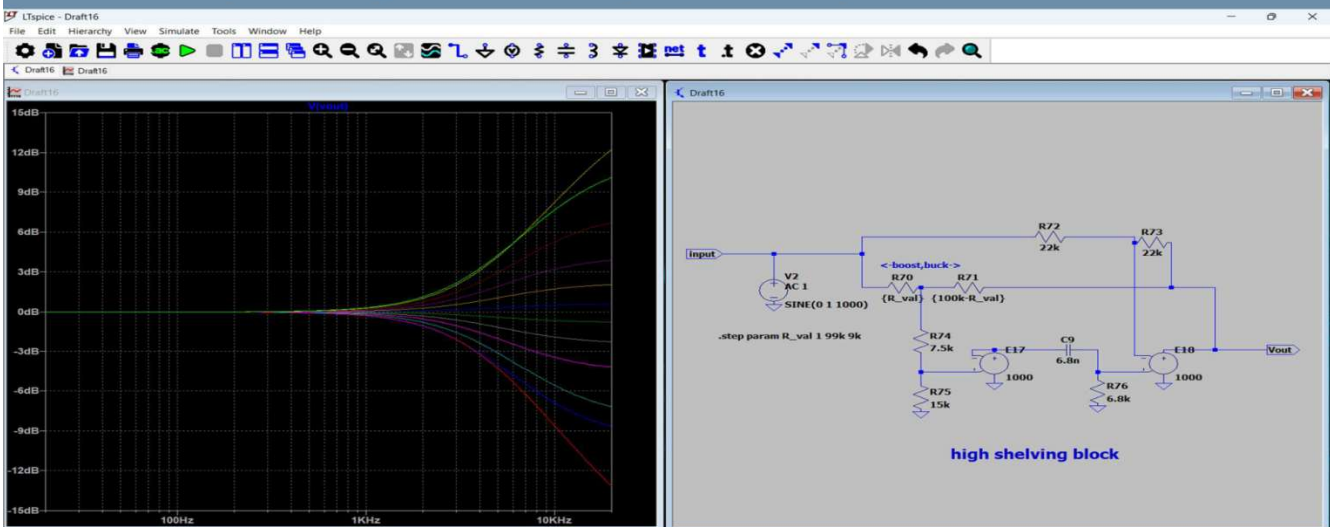
Block Analysis — High Shelf Filter (frequency control)



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Figure 4.7 — High shelf filter: corner frequency shift with RV15 (LTspice)

Block Analysis — High Shelf Filter (Gain Control)



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Figure 4.8 — High shelf filter: gain sweep -9 dB to +9 dB (LTspice)



5. Input/Output Interface Design

5.1 Input Stage — 3.5 mm TRRS Jack

J1 is a 4-pole TRRS (Tip-Ring-Ring-Sleeve) 3.5 mm female audio jack. The TRRS pinout follows the CTIA standard: Tip = Left audio, Ring 1 = Right audio, Ring 2 = Microphone, Sleeve = Ground. The input accepts audio from any consumer device — smartphone, laptop, tablet, or portable media player.

A 10 μF electrolytic DC-blocking capacitor (C1) in series removes any DC bias present at the source output. The signal then passes through a voltage divider formed by R1 and R2 (both 10 k Ω), presenting an input impedance of approximately 5 k Ω to the source. This is sufficiently high to avoid loading a standard headphone output (32–100 Ω source impedance) while attenuating the signal to an op-amp-compatible level.

5.2 ESD Protection — TVS1400DRV

TVS1400DRV transient voltage suppressor arrays (U1 at input, U2 at output) clamp ESD transients to safe levels. These bidirectional devices protect the TL074CDR op-amp inputs and the user's source device against electrostatic discharge events that occur during hot-plugging of audio connectors. Placement is immediately adjacent to the audio jack footprints to minimise the unprotected trace length.

5.3 Output Stage — 3.5 mm TRRS Jack

J2 is a second 4-pole TRRS 3.5 mm female jack providing the equalizer output. The signal is buffered by the final TL074 stage. R4 (100 Ω) in series with the output ensures stability when driving capacitive loads such as headphone cables (typically 100–400 pF). The output can directly drive 32 Ω headphones from the op-amp output swing, or it can be connected to the input of an external power amplifier for loudspeaker use. C5 (100 μF) and R24 (47 Ω) form a supply decoupling filter for this output section.

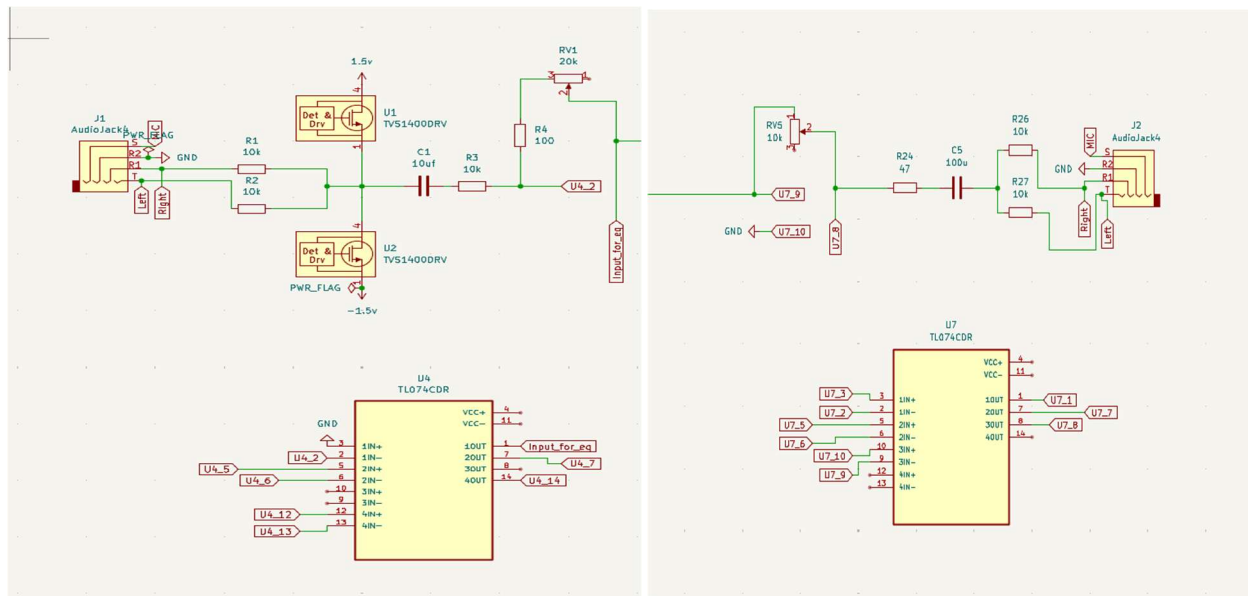


Figure 5.1 — KiCad schematic: input(left) and output(right) interface stage.



6. Bill of Materials

All components listed below are as captured in the KiCad schematic (version 9.0). Resistors are 1% tolerance metal-film; capacitors are polyester/polypropylene film type unless noted. Total component count: 89 footprints across 368 nets.

6.1 Integrated Circuits

Ref.	Part Number	Description	Qty
U3–U7	TL074CDR	Quad JFET-input op-amp, SOIC-14	5
U1, U2	TVS1400DRV	TVS ESD protection diode array	2

6.2 Fixed Resistors

Ref.	Value	Function
R1, R2	10 k Ω	Input voltage divider — impedance matching
R3, R36, R48	10 k Ω	SVF summing node resistors (Bands 1–3)
R4	100 Ω	Output series resistor — headphone load stability
R5, R8, R10, R15, R17–R22, R30–R35, R42–R47, R54, R56	22 k Ω	SVF feedback and summing network
R6, R52	7.5 k Ω	Shelf filter stability resistors
R7	15 k Ω	Low shelf stability
R9, R55	6.8 k Ω	Shelf integrator input resistors
R11, R16, R38, R50	5 k Ω	SVF output network
R12, R29, R41	3.3 k Ω	SVF input resistors
R13, R28, R40	1 k Ω	SVF feedback scaling
R14, R17, R26, R27, R39, R51	10 k Ω	SVF node / summing resistors
R24	47 Ω	Power supply decoupling
R53	3.7 k Ω	High shelf stability

6.3 Potentiometers (Controls)

Ref.	Value	Function
RV1	20 k Ω	Low shelf gain control (boost / cut ± 9 dB)
RV2	100 k Ω dual gang	Low shelf corner frequency (120–500 Hz)
RV3, RV4	10 k Ω dual gang	Band 1 center frequency (120–500 Hz)



Ref.	Value	Function
RV5	10 k Ω	Band 1 gain control (± 9 dB)
RV7, RV9	10 k Ω dual gang	Band 2 center frequency (400–1600 Hz)
RV8	10 k Ω	Band 2 gain control (± 9 dB)
RV10, RV12	10 k Ω dual gang	Band 3 center frequency (1.2–5 kHz)
RV11	10 k Ω	Band 3 gain control (± 9 dB)
RV13, RV14	10 k Ω	Q control (bandwidth) for Bands 1 and 2
RV15	100 k Ω dual gang	High shelf corner frequency (1.2–5 kHz)
RV16	10 k Ω	High shelf gain control (boost / cut ± 9 dB)

6.4 Capacitors

Ref.	Value	Type / Function
C1	10 μ F	Electrolytic — input DC blocking
C2	6.8 nF	Film — high shelf integrator
C3, C4	33 nF	Film — Band 1 (250 Hz) integrators ($\times 2$)
C5	100 μ F	Electrolytic — output power supply decoupling
C7, C8	10 nF	Film — Band 2 (800 Hz) integrators ($\times 2$)
C9, C10	3.3 nF	Film — Band 3 (2.5 kHz) integrators ($\times 2$)
C11	80 nF	Film — low shelf integrator

6.5 Connectors

Ref.	Part	Description
J1	AudioJack4 (TRRS 3.5 mm, female)	Stereo input from phone / laptop / any consumer source
J2	AudioJack4 (TRRS 3.5 mm, female)	Stereo output to headphones or power amplifier input



7. KiCad Schematic and PCB Design

7.1 Schematic Capture — KiCad EESchema

The complete circuit was formally captured as a schematic in KiCad EESchema (version 9.0). The schematic encompasses all five filter blocks, the input/output interface, power supply connections, TVS protection, and decoupling networks.

Parameter	Value
Tool	KiCad EESchema v9.0
Total components (BOM)	89 footprints
Total nets	368
Op-amp ICs (TL074CDR)	5 (U3–U7) — 20 op-amp units total
Potentiometers	16 (RV1–RV16)
Fixed resistors	56
Capacitors	11
Audio connectors	2 (J1, J2 — TRRS 3.5 mm)
TVS protection ICs	2 (U1, U2)

7.2 PCB Layout — KiCad PCBnew

Following schematic capture, a full PCB layout was completed in KiCad PCBnew (version 9.0). A 4-layer stackup was chosen to provide dedicated power and ground planes, reducing supply noise and crosstalk between the sensitive analog filter stages.

7.2.1 Layer Stackup

Layer	Name	Role
Layer 1 (Top)	F. Cu	Primary signal routing, all component placement (SMD)
Layer 2 (Inner 1)	In1.Cu	Solid GND plane — continuous low-impedance return path
Layer 3 (Inner 2)	In2.Cu	Power plane — ± 15 V rail distribution
Layer 4 (Bottom)	B. Cu	Secondary signal routing

7.2.2 PCB Statistics

PCB Parameter	Specification
PCB tool	KiCad PCBnew v9.0
Board stackup	4-layer FR-4



PCB Parameter	Specification
Total board thickness	1.786 mm
Copper — outer layers	43 μm (F. Cu, B. Cu)
Copper — inner layers	35 μm (In1.Cu, In2.Cu)
Solder mask	Both sides ($\epsilon_r = 3.5$)
Silk screen	Both sides (component ref. labels)
Total track segments	1,468
Total vias	150
Drawing sheet	A3

7.2.3 Layout Strategy

- Filter stages are arranged in signal-flow order (left to right) to minimise long return paths.
- Decoupling capacitors (C5, C11) are placed immediately adjacent to op-amp power supply pins.
- Audio input and output connectors (J1, J2) are placed on opposite edges of the board for clean front-panel mounting.
- TVS protection ICs (U1, U2) are located directly adjacent to audio jack footprints, minimising the unprotected trace length.
- All filter capacitors (C2–C4, C7–C10) are placed within 5 mm of their associated op-amp integrator stage.
- Potentiometers (RV1–RV16) are arranged along one edge of the board for front-panel mounting through an enclosure.
- The inner GND plane (In1.Cu) provides a continuous low-impedance return beneath all op-amp stages, suppressing common-mode interference.

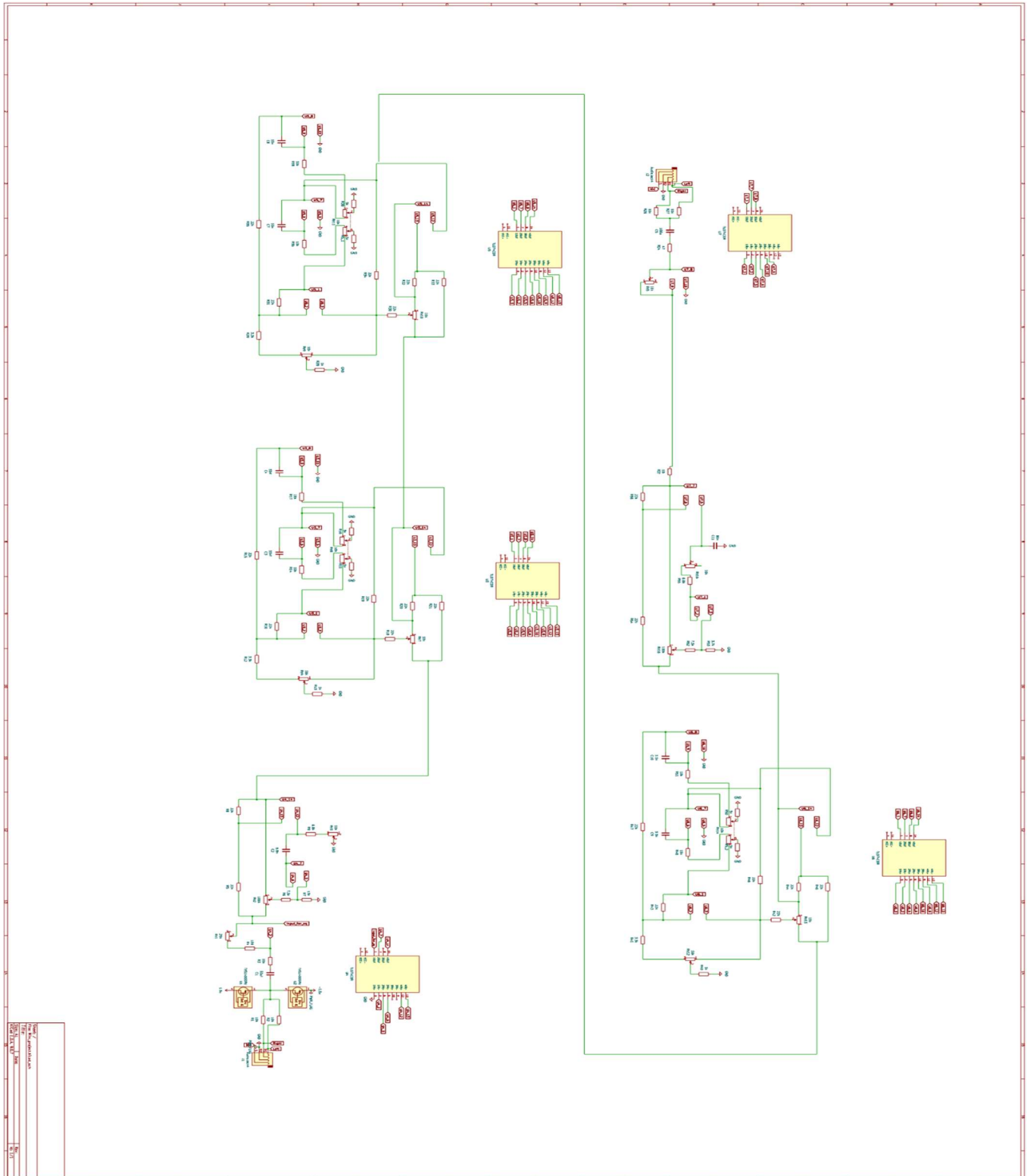


Figure 7.1 — Complete KiCad EESchema schematic of the 5-band analog equalize

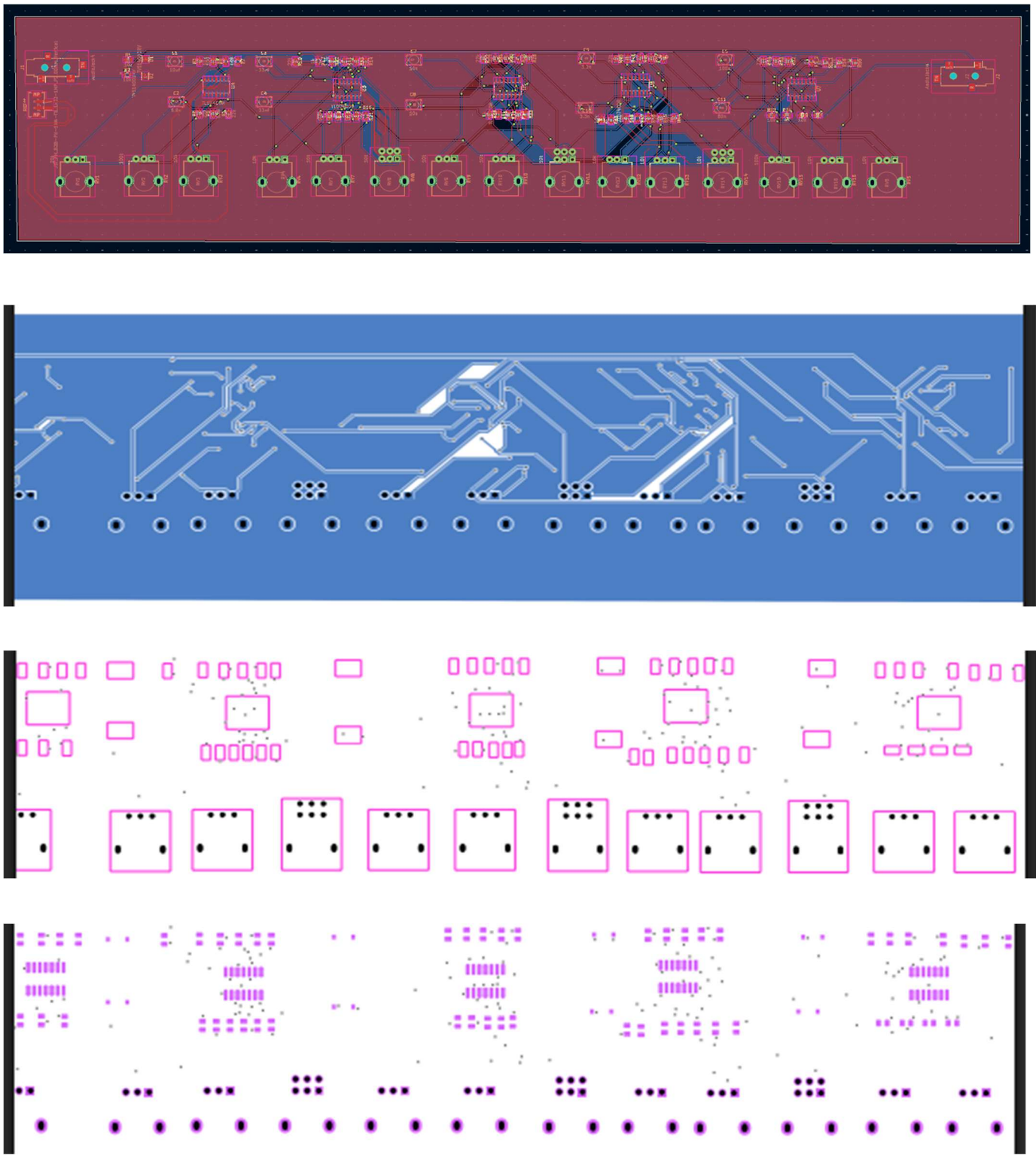


Figure 7.3 — KiCad PCBnew: different layers showing component placement and routing.

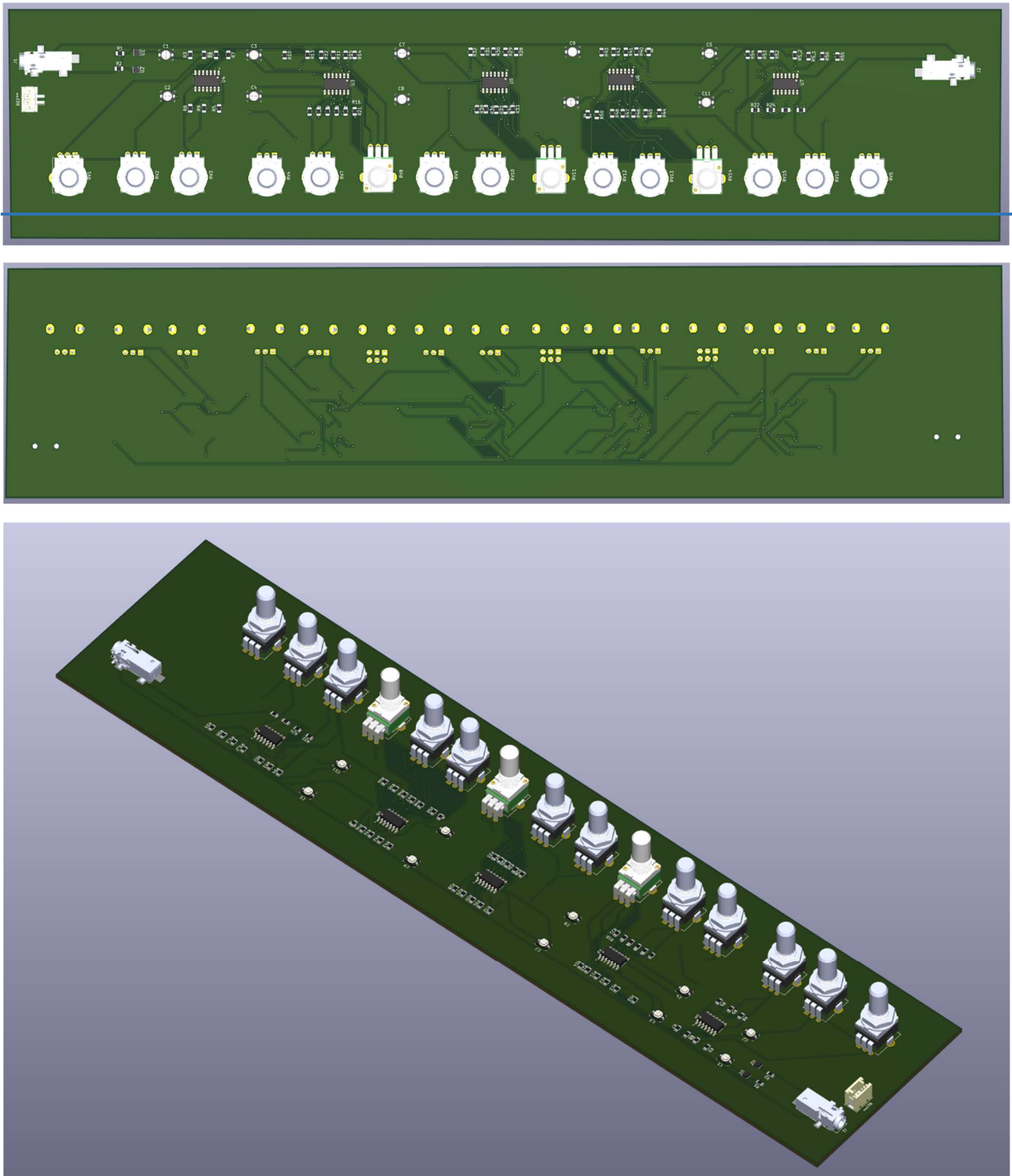


Figure 7.4 — KiCad 3D view (top, bottom, isometric view) of the assembled PCB



8. Simulation Results

8.1 LTspice AC Sweep — Individual Block Verification

Each filter block was independently simulated in LTspice XVII using AC analysis from 20 Hz to 20 kHz (1000 points per decade). Initial verification used behavioral VCVS (Voltage Controlled Voltage Source) models to confirm topology and transfer function behaviour. These were subsequently replaced with the exact TL074CDR SPICE model for final verification.

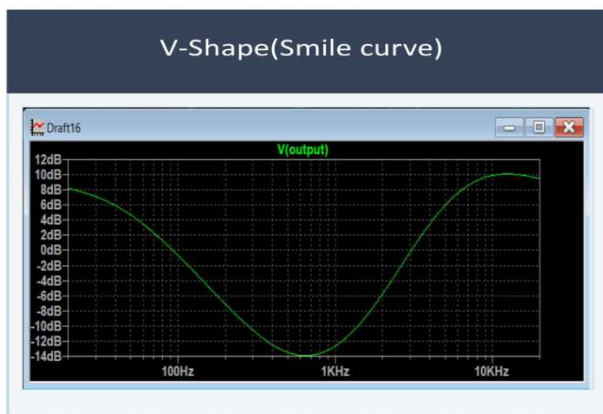
Block	Target f_0 / f_c	Simulated Result	Gain Range Achieved	Status
Low Shelf	250 Hz nominal	~250 Hz, tunable 120–500 Hz	–9 to +9 dB	✓ Pass
Band 1 (Bass)	250 Hz	~250 Hz, tunable 120–500 Hz	–9 to +9 dB	✓ Pass
Band 2 (Midrange)	800 Hz	~800 Hz, tunable 400–1600 Hz	–9 to +9 dB	✓ Pass
Band 3 (Upper Mid)	2.5 kHz	~2.5 kHz, tunable 1.2–5 kHz	–9 to +9 dB	✓ Pass
High Shelf	2.3 kHz nominal	~2.3 kHz, tunable 1.2–5 kHz	–9 to +9 dB	✓ Pass

8.2 Preset EQ Response Verification

The complete cascaded design was simulated with popular equalizer preset settings. Results were verified via LTspice AC sweep across the full audio band.

Simulation Results — Some popular Equalizer setting

.ac dec 1000 20 20K — LTspice AC Sweep



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Figure 8.1 — LTspice: V-Shape (Smile Curve) response [left] and Vocal Boost response [right]



- V-Shape (Smile Curve): Low shelf and high shelf at +9 dB, all three peaking bands at –9 dB. Result: bass and treble boosted, midrange attenuated — characteristic smile-curve response confirmed.
- Vocal Boost: Band 2 (800 Hz) and Band 3 (2.5 kHz) at moderate boost (+4 to +5 dB), Band 1 at slight cut, shelves at moderate settings. Result: enhanced midrange presence and vocal intelligibility confirmed.

8.3 Proteus Verification — Exact Component Models

Following LTspice verification, the complete circuit was simulated in Proteus Design Suite using the exact TL074C SPICE model replacing all behavioral sources. Simulation results confirmed that the design meets all specifications under realistic device parameters:

- Frequency response: All five bands achieved target center frequencies and gain ranges, matching LTspice predictions within $\pm 2\%$.
- Slew rate: The TL074C slew rate of 13 V/ μ s was confirmed adequate — no slew-induced distortion observed at 20 kHz full-scale.
- Op-amp output swing: Confirmed ± 13 V output swing under ± 15 V supply, providing adequate headroom for audio levels.
- DC offset: Measured output DC offset < 5 mV across all stages — negligible for audio applications.

Overall, Proteus simulation results match LTspice predictions, validating the design and confirming readiness for PCB fabrication.

9. Challenges Faced and Solutions

Challenge	Problem	Solution
Q-Gain Coupling	In standard SVF, BP peak = $Q \times V_{in}$. Tuning Q inadvertently changes boost/cut magnitude.	Q-control resistor relocated to the BP output feedback path of the summing amplifier. Peak amplitude is now constant at V_{in} , fully decoupled from Q.
Flat Response at Centre Pot	At the mid-position of the gain pot, the dry and wet signal paths must exactly cancel for a true 0 dB response. Resistor mismatch causes residual gain error.	Gain network resistors (R11, R16, R28, R29, R38, R50) carefully ratio-matched in simulation to achieve symmetric ± 9 dB swing centred on 0 dB.
Frequency Control Coupling	Changing only one integrator's resistance breaks SVF loop symmetry, distorting filter shape and shifting effective Q.	Dual-gang potentiometers used for all frequency controls (RV2, RV3/RV4, RV7/RV9, RV10/RV12, RV15) — both integrators always track equal RC time constants.
Shelf Filter Precision	Conventional Baxandall tone controls have imprecise f_c , gain-frequency interaction,	Replaced with SVF-based shelves — LP output (low shelf) and HP output (high shelf) each



Challenge	Problem	Solution
	and limited independent control.	summed with the dry signal, giving full independent gain and frequency control.
Input Impedance Mismatch	Consumer source devices have low-impedance headphone outputs. A too-low load causes current limiting and distortion at the source.	10 μ F DC-blocking capacitor + 10 k Ω voltage divider (R1, R2) presents ~5 k Ω input impedance — safe for all consumer audio outputs.
ESD at Audio Connectors	Hot-plugging 3.5 mm jacks generates transients up to several hundred volts, capable of destroying op-amp input stages.	TVS1400DRV arrays (U1, U2) at both jacks clamp transients to safe levels before they reach any active device.

10. Future Scope

10.1 Hardware Fabrication and Assembly

The PCB design will be sent for fabrication. Assembly will use TL074CDR op-amps and all components specified in the BOM (Section 6) on ± 15 V regulated supply rails. Initial functional testing will use a signal generator as the audio source and an oscilloscope or audio analyser to measure the frequency response at each stage output.

10.2 Audio Characterisation

The assembled board will be characterised using real audio signals — pink noise, music, and swept sine tones. Frequency response, THD+N (total harmonic distortion plus noise), dynamic range, and noise floor will be measured using audio analysis software such as REW (Room EQ Wizard) and compared against simulation predictions to fully validate the design.

10.3 PCB Noise Optimisation

Based on measured results, component values and PCB layout may be refined to reduce noise floor. Attention will be given to supply decoupling placement, ground loop elimination, and shielding of high-gain stages.

10.4 Enclosure and Panel Design

A suitable enclosure will be designed with front-panel cutouts for all 16 potentiometers and both 3.5 mm audio jacks, producing a standalone consumer-grade audio device.

10.5 Extended Band Count

As a future extension, the architecture can be expanded to 7 or 10 bands by adding additional SVF peaking stages. An input buffer/preamp and output makeup gain amplifier would also be incorporated for professional-grade signal handling.



11. Conclusion

This report presents the complete design and implementation of a 5-band analog parametric audio equalizer. Starting from a theoretical foundation in State Variable Filter (KHN) topology and audio frequency perception, the project progressed through analytical transfer function derivation, behavioral VCVS simulation in LTspice, replacement with exact TL074CDR op-amp models, formal schematic capture in KiCad EESchema, and complete 4-layer PCB layout in KiCad PCBnew.

The five filter blocks — a low shelving filter, three tunable SVF peaking bands at 250 Hz, 800 Hz, and 2.5 kHz, and a high shelving filter — each provide ± 9 dB of gain control and tunable center/corner frequency via dual-gang potentiometers. A key design achievement is the Q-gain decoupled architecture, where bandwidth and boost/cut magnitude are fully independent. The addition of a proper input/output interface with 3.5 mm TRRS jacks, TVS ESD protection, and headphone-compatible output completes the transition from a laboratory circuit to a self-contained, usable audio device.

Simulation in both LTspice and Proteus using exact component models confirmed that all specifications are met across the full 20 Hz – 20 kHz audio band. The KiCad schematic (89 components, 368 nets) and 4-layer PCB layout (1,468 track segments, 150 vias) represent a complete, fabrication-ready design. Remaining steps — PCB fabrication, assembly, and audio-domain characterisation — will finalise the project as a working hardware prototype.

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